

Benjamin Wrinn

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EDUCATION

University of Pennsylvania - The School of Engineering and Applied Science

Philadelphia, PA

Candidate for BSE in Mechanical Engineering

May 2029

- **Courses:** Statics & Strength of Materials, Intro to Thermal Fluids, Linear Algebra, Calculus 1-3, Physics 1&2, Chemistry, Python

Oregon Episcopal School - High School

Portland, OR

- *Cumulative GPA:* 3.9/4.0 (UW); SAT: 1570 | Math 780 | Writing 790

May 2022

RELEVANT EXPERIENCE

Vero Electric

Philadelphia, PA

Lead Mechanical Engineer

June 2026 – Present

- Engineered a custom, **261kWh/135kW BESS enclosure** from Solidworks to physical assembly, designing for -20 - 55°C ambient extremes and compliance with UL safety standards. (Certification by end of 2026)
- Optimized for low-cost manufacturing; validated architecture through Solidworks **Thermal** and **Static** simulations
- Routed a 64-cell-series **BMS node PCB** in KiCad compatible with industry-leading battery manufacturers, cutting node integration time by enabling drop-in fit with existing pack architecture

Penn Electric Racing Formula SAE EV Team

Philadelphia, PA

Mechanical Team

Sept 2025 – Present

- Designing **brake-absorption dynamometer** in preparation for 4WD and Design Competition, that will later be converted for *regen brake* testing that withstands loads of 500nM torque and 20,000 RPM
- Engineering a servo-actuated Drag Reduction System projected to cut straight-line drag by **37.7%**, improving lap times by **2 seconds**
- Produced KeyShot renders used in sponsor and design-review presentations. Modeled and drew chassis components in SolidWorks per GD&T standards. Manufactured carbon fiber parts via resin infusion at sponsor facilities

American Rocketry Challenge

Portland, OR

Team Host, Hardware and Media Lead

Sept 2022 – August 2025

- Engineered a dynamic air brake system that autonomously adjusted drag using a Python lookup table, landing the rocket **within 7ft of target apogee**: placing **1st** in State, **20th** nationally and **Top 5** in Presentation
- Designed the control PCB and validated drag predictions using CFD (Solidworks) before flight testing
- Grew team Instagram by +5.5k followers and +350k interactions as Media Lead, driving sponsor visibility and recruitment

Children's Cancer Therapy Development Institute

Portland, OR

Student Research Assistant

June 2023 – June 2025

- Examined feasibility of quail embryo drug trials as manuscript co-author (pending publication); Wet-lab experience in both CAM and Bioluminescence assays
- Fabricated a prototype for machine vision on quail embryo drug trials, using OpenCV to quantify embryo health; cc-TDI has continued to develop this as an economical screening method to pair with mouse clinical trials.
- Filmed and edited quail instrument instruction videos for customers

ADDITIONAL

- **Skills:** CAD (SolidWorks – parts, assemblies, drawings, GD&T), **Manufacturing** (CNC Machining, 3D Printing, Laser Cutting, Wet Layups), **Simulations** (ANSYS, Star CCM+, SolidWorks FEA, Luminary Cloud), **Kicad**/Altium PCB Design, **Python**, Blender, Davinci Resolve, Chinese (Mandarin), Completed *NFPE 70E* Training
- **Activities:** Weightlifting, USA National Kung Fu Team (Gold, Silver Medalist @ 2023 IWUF, Youth Coach), William Temple House Food Pantry Volunteer (stocked 3700+ lb of pantry items, fulfilled 120+ orders, coded menu translator in Python)